

NAGAR DRISHTI (नगर दृष्टि)

Autonomous Edge-AI Urban Intelligence & Municipal Infrastructure Platform

SIH 2026 PITCH SCRIPT

🕒 4 Speakers • 6-7 Min
Total

SPEAKER	CORE FOCUS AREA	KEY MESSAGE & NARRATIVE GOAL	EST. TIME
Speaker 1	Problem & Solution	The paradox of modern India; reactive complaints vs. transit-mounted fleet scanning.	90s
Speaker 2	Technical Approach	Edge perception (YOLO-v8, 30 FPS, <25ms), 3D volumetric math, spatial deduplication.	90s
Speaker 3	Feasibility & Viability	Zero-Capex transit synergy, open civic APIs (CP-GRAMS/ICCC), 30-40% budget savings.	90s
Speaker 4	Impact & Grand Finale	Saving 3,500+ lives, Viksit Bharat 2047 vision, memorable patriotic punchline.	90s

SPEAKER 1 — The Problem & The Paradigm Shift

TIME: 00:00 - 01:30 (90s)

Stage Presence: Step firmly to center stage. Confident, empathetic eye contact with judges. Set an urgent, relatable tone.

"Respected jury members — consider the paradox of modern India. We are sending rovers to the Moon, yet every day millions navigate roads with dangerous 15-centimeter potholes, dark avenues under dangling live wires, flooded corridors from burst water pipes, and overflowing sewer lines.

Why? Because municipal governance today is **painfully reactive**. Cities rely on citizen grievance apps where complaints take 2 to 3 weeks to be verified, or on sporadic manual surveys that cost crores and are outdated before ink dries.

We asked: **What if our cities could inspect themselves automatically every hour, without adding a single extra patrol vehicle?**

Meet **Nagar Drishti (नगर दृष्टि)**. We transform existing public transit fleets — buses that already travel every arterial road daily — into an active, intelligent mobile scanner network. With custom Edge AI cameras scanning roads, streetlights, water mains, and sanitation, defects are verified and converted into engineering work orders in seconds."

Handoff Cue: "To explain the deep-tech engineering powering this platform, I hand over to [Speaker 2's Name]."

***Stage Presence:** Step forward briskly. Point toward the architecture slide or laptop screen. Crisp, authoritative engineering voice.*

"Thank you, [Speaker 1]. Behind Nagar Drishti is an **Edge-First, Three-Tier Architecture**:

- 1. Multi-Modal Edge Perception:** Each bus carries an industrial-grade Edge AI processor linked to Front, Rear, and Side cameras. Running optimized, quantized **YOLO-v8 models at >30 FPS with sub-25ms inference**, we don't just detect defects — our dual-frame geometry calculates **precise 3D dimensions**: pothole crater volume in m^3 , streetlight pole deflection angles, water burst flow rates ($\sim 800\text{L/min}$), and sewer overflow spread radius.
- 2. Privacy-by-Design & Zero Bandwidth Waste:** No continuous video is streamed. Face and license-plate blurring happen on-chip. Only verified 10-second incident packages with GPS telemetry sync to the cloud.
- 3. Automated Engineering Work Orders:** Our cloud core performs **spatial deduplication** — 5 buses seeing the same pothole generate only 1 ticket. It then synthesizes exact materials bills (bitumen hot-mix, dewatering pumps) and auto-routes tickets to PWD, Electrical, or Water departments under strict SLA timers."

Handoff Cue: "Now, how practical and economically viable is this for Indian municipalities? For that, here is [Speaker 3's Name]."

SPEAKER 3 — Feasibility, Scalability & Viability

TIME: 03:00 – 04:30 (90s)

Stage Presence: Pragmatic, commercial-mindset. Use deliberate hand gestures when discussing cost comparisons and percentages.

"Thank you, [Speaker 2]. Whenever AI is proposed to government bodies, the question is: *'Who pays for it, and will it work on the ground?'* Nagar Drishti is designed for **frugal, high-yield Indian municipal economics**:

- 1. Zero Infrastructure Capex:** Static smart-poles cost lakhs each and have 50-meter blind spots. Meanwhile, Delhi alone runs 7,000 transit buses; Bengaluru runs 6,500. By equipping vehicles already burning fuel and covering routes, edge camera retrofits cost under **₹15,000 per bus**, delivering **98% arterial road coverage every 24 hours**.
- 2. Seamless Institutional Synergy:** Nagar Drishti integrates smoothly with existing command centers (ICCC) and grievance backends (CP-GRAMS) via RESTful APIs. When repairs finish, the very next bus passing that GPS point automatically re-scans the surface to close the SLA loop without human bias.
- 3. Massive ROI:** Catching a 15cm pothole before it destroys the road base, or a leaking pipe before it creates a sinkhole, shifts cities from costly reconstruction to affordable preventative maintenance — saving cities an estimated **30% to 40% in annual maintenance budgets**."

Handoff Cue: "To summarize our human impact and long-term vision, I invite **[Speaker 4's Name]**."

SPEAKER 4 — Impacts, Benefits & The SIH 2026 Finale

TIME: 04:30 – 06:00 (90s)

Stage Presence: Step forward with heartfelt conviction. All 4 teammates stand shoulder-to-shoulder in a confident, unified posture.

"Thank you, [Speaker 3]. At Nagar Drishti, our ultimate metric isn't code or compute — it is **human lives and public safety**:

- **Zero-Death Vision:** Over 3,500 Indians lose their lives to road defects annually. By slashing detection time from **21 days to under 15 minutes**, we eliminate road hazards before they claim lives.
- **Contractor Accountability:** Cryptographically timestamped before-and-after video audits eliminate ghost repairs and substandard patches.
- **Viksit Bharat 2047:** As India builds world-class infrastructure, smart cities cannot run on 20th-century paper complaints. A true smart city must have eyes that observe, a mind that understands, and hands that repair."

[Pause for 2 seconds. Look directly into the eyes of the judges. Tone becomes calm, steady, and inspiring.]

"Honorable judges — with every wheel that turns on our city streets, Nagar Drishti transforms our buses into tireless guardians of our public infrastructure.

Together, let us build an India where our roads are safe, our cities are responsive, and no citizen's voice is needed to fix what our technology should have seen first.

Thank you, and Jai Hind! IN"



QUICK JURY Q&A DEFENSE CHEAT SHEET

Q: Poor night lighting or monsoon rains?

Speaker 2: "Models trained on dusk, headlights & wet reflections with CLAHE adaptive contrast. Live tests confirm >92% confidence."

Q: What if cellular connectivity drops?

Speaker 2: "Full Edge autonomy. Inference runs offline on NVMe storage; incident packages sync automatically upon reaching depot Wi-Fi/5G."

Q: Who maintains the bus cameras?

Speaker 3: "Transit fleets undergo nightly depot washing and maintenance. Nodes have auto-health diagnostics powered by bus battery step-down."

Q: How do contractors adopt this?

Speaker 3: "They get turn-by-turn navigation, video evidence, and required materials sheets. The next passing bus autonomously verifies completion."